

[0116] The cyclic hydrocarbon group for Ra^{01} and Ra^{02} may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group, and may be polycyclic or monocyclic.

[0117] As the monocyclic aliphatic hydrocarbon group, a group in which 1 hydrogen atom has been removed from a monocycloalkane is preferable. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

[0118] As the polycyclic aliphatic hydrocarbon group, a group in which 1 hydrogen atom has been removed from a polycycloalkane is preferable, and the polycyclic group preferably has 7 to 12 carbon atoms. Examples of the polycycloalkane include adamantane, norbornane, isobornane, tricyclodecane and tetracyclododecane.

[0119] The aromatic hydrocarbon group for Ra^{01} and Ra^{02} is a hydrocarbon group having at least one aromatic ring. The aromatic ring is not particularly limited, as long as it is a cyclic conjugated compound having $(4n+2)$ π electrons, and may be either monocyclic or polycyclic. The aromatic ring preferably has 5 to 30 carbon atoms, more preferably 5 to 20, still more preferably 6 to 15, and most preferably 6 to 12. Examples of the aromatic ring include aromatic hydrocarbon rings, such as benzene, naphthalene, anthracene and phenanthrene; and aromatic hetero rings in which part of the carbon atoms constituting the aforementioned aromatic hydrocarbon rings has been substituted with a hetero atom. Examples of the hetero atom within the aromatic hetero rings include an oxygen atom, a sulfur atom and a nitrogen atom. Specific examples of the aromatic hetero ring include a pyridine ring and a thiophene ring. Specific examples of the aromatic hydrocarbon group include a group in which one hydrogen atom has been removed from the aforementioned aromatic hydrocarbon ring or aromatic hetero ring (aryl group or heteroaryl group); a group in which one hydrogen atom has been removed from an aromatic compound having two or more aromatic rings (biphenyl, fluorene or the like); and a group in which one hydrogen atom of the aforementioned aromatic hydrocarbon ring or aromatic hetero ring has been substituted with an alkylene group (an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group). The alkylene group bonded to the aforementioned aromatic hydrocarbon ring or the aromatic hetero ring preferably has 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and most preferably 1 carbon atom.

[0120] In the case where the hydrocarbon group for Ra^{01} and Ra^{02} is substituted, examples of the substituent include a hydroxy group, a carboxy group, a halogen atom (such as a fluorine atom, a chlorine atom or a bromine atom), an alkoxy group (such as a methoxy group, an ethoxy group, a propoxy group or a butoxy group), and an alkyloxycarbonyl group.

[0121] In formula (a0-r1-1), Ra^{01} and Ra^{02} may be mutually bonded to form a ring.

[0122] Examples of the ring structure formed by Ra^{01} and Ra^{02} include an alicyclic hydrocarbon group, and a condensed ring of an alicyclic group and an aromatic hydrocarbon group. The ring structure formed by Ra^{01} and Ra^{02} may have a hetero atom.

[0123] The alicyclic hydrocarbon group formed by Ra^{01} and Ra^{02} may be either a polycyclic group or a monocyclic group.

[0124] As the monocyclic aliphatic hydrocarbon group, a group in which 1 hydrogen atom has been removed from a monocycloalkane is preferable. The monocycloalkane preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

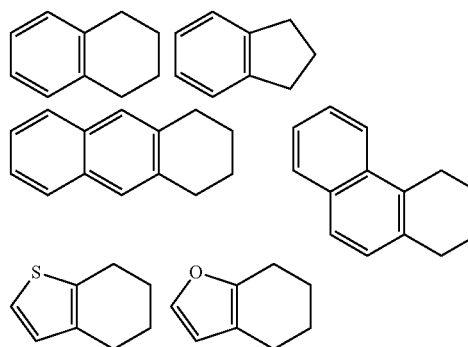
[0125] As the polycyclic aliphatic hydrocarbon group, a group in which 1 hydrogen atom has been removed from a polycycloalkane is preferable, and the polycyclic group preferably has 7 to 12 carbon atoms. Examples of the polycycloalkane include adamantane, norbornane, isobornane, tricyclodecane and tetracyclododecane.

[0126] In the condensed ring formed by Ra^{01} and Ra^{02} , the alicyclic hydrocarbon portion may be monocyclic or polycyclic, and the aromatic hydrocarbon portion may be monocyclic or polycyclic.

[0127] The condensed ring may have a substituent. Examples of the substituent include a methyl group, an ethyl group, a propyl group, a hydroxy group, a hydroxyalkyl group, a carboxy group, a halogen atom (such as a fluorine atom, a chlorine atom or bromine atom), an alkoxy group (such as a methoxy group, an ethoxy group, a propoxy group or a butoxy group), an acyl group, an alkyloxycarbonyl group, and an alkylcarbonyloxy group.

[0128] Specific examples of the condensed ring formed by Ra^{01} and Ra^{02} are shown below.

[Chemical Formula 7]



[0129] In formula (a0-r1-1), in terms of improving the etching resistance, it is preferable that Ra^{01} and Ra^{02} are mutually bonded to form a ring, and it is more preferable that Ra^{01} and Ra^{02} are mutually bonded to form an alicyclic hydrocarbon group.

[0130] In formula (a0-r1-1), Y_a^0 represents a quaternary carbon atom. That is, the number of carbon atoms bonded to Y_a^0 (carbon atom) is 4.

[0131] In formula (a0-r1-1), Ra^{031} , Ra^{032} and Ra^{033} each independently represents a hydrocarbon group which may have a substituent.

[0132] The hydrocarbon group (which may have a substituent) for Ra^{031} , Ra^{032} and Ra^{033} is the same as defined for the hydrocarbon group (which may have a substituent) for Ra^{01} and Ra^{02} in the aforementioned formula (a0-r1-1). Among these examples, as the hydrocarbon group (which may have a substituent) for Ra^{031} , Ra^{032} and Ra^{033} , a linear or branched alkyl group which may have a substituent is preferable, and a linear alkyl group is more preferable.

[0133] However, at least one of Ra^{031} , Ra^{032} and Ra^{033} is a hydrocarbon group having a polar group.